

Adversarial Machine Learning

Homework Assignment 5

The assignment is due by the end of the day on Thursday, April 17.

Please note that the submitted solutions are expected to follow the examples specified below that are posted on Canvas. If your code does not follow the provided examples, marks will be deducted. Also, marks will not be assigned if the codes are generated by coding assistants. To repeat, it is okay if you write the code and afterward use help from coding assistants to debug the code: it is not okay to ask a coding assistant to generate the code and then copy-paste it into the notebook.

And to remind you again, the assignments and solutions from the offering of the AML course in the previous semesters can be found at these links: Spring 2023 ([link](#)), and Fall 2021 ([link](#)). Also, if you need to refresh your knowledge about training neural networks, the materials from the Applied Data Science with Python course can be found [here](#).

Objective:

- Implement a backdoor poisoning attack against deep learning classification models.
- Implement adversarial attacks against ML classification models used in cybersecurity applications.

Part 1: Backdoor Poisoning Attack (50 marks)

The Backdoor Attack in this assignment is based on the BadNet paper by Gu et al. (2019), which we covered in Lecture 8. The notebook `Backdoor_Attack` in the Codes folder on Canvas provides an example of implementing this attack. **It is expected that your solution follows the example provided in the `Backdoor_Attack` notebook.** In addition, the following [notebook](#) in the Adversarial Robustness Toolbox explains the implementation of the boundary attack on MNIST images.

Dataset: We will use the SVHN (Street View House Numbers) dataset. The dataset consists of about 48,000 images of house numbers from street views. We will use a portion of the cropped version of the images in the dataset. Examples are shown in Figure 1. You can notice that the label of the images refers to the middle digit, and there are other distracting digits on the sides. The size of the images is 32×32 pixels. More information about the dataset can be found at this [link](#).

Task 1: Step 1. Load the dataset and plot a figure with at least 9 images and the corresponding ground-truth labels. Note that the labels are the digits of the images. The compressed folder with the images `SVHN_dataset.zip` and the file `SVHN_labels.csv` having the labels are provided in the assignment folder.

Step 2. Train a **VGG-16** convolutional neural network for classification of the house numbers images in the dataset **using the TensorFlow library**.

Please use 20% of the data for a test set, and another 20% from the remaining images for a validation set, similar to the data splits we used in the previous assignments.

Use the notebook Backdoor_Attack in the Codes folder on Canvas as guidance for loading the dataset, and training and evaluating the model.

Perform hyperparameter tuning to obtain accuracy on the test dataset above 93%.

Estimated running time: between 5 and 20 minutes.

Report (20 marks): (a) Report the classification accuracy for the train set, validation set, and test set of images. For full marks, it is expected to report test accuracy above 93%. (b) Plot the training and validation loss and accuracy curves. If applicable, provide any other observations regarding the training of the model.



Figure 1. Sample images from the SVHN dataset.

Task 2: The goal of the backdoor attack is to misclassify poisoned images of the digit 2 with the backdoor pattern as digit 5 images. Therefore, the poisoned model should have high classification accuracy on clean images without a backdoor pattern, as well as high classification accuracy on digit 2 images with a backdoor pattern that are predicted to be digit 5 images.

Use the notebook Backdoor_Attack in the Codes folder on Canvas as guidance for implementing the attack.

Step 1: Create a backdoor trigger pattern for poisoning the model. Use the following pattern shown in Figure 2, which consists of 4 white pixels located in the lower-right corner of the images. Include in your report a plot of an example image with a backdoor trigger.

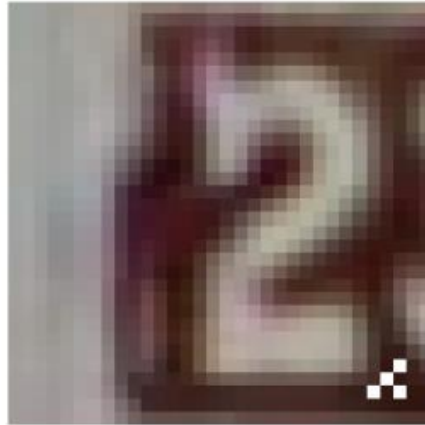


Figure 2. The pattern of 4-pixels for the backdoor attack.

Step 2: Create a poisoned training dataset, by using the training and validation sets of SVHN images. Use 20% of the total number of images of digit 2, and apply the backdoor pattern to create poisoned images for training. Print the size of the training dataset and the number of poisoned images. Plot at least 9 images with the applied backdoor pattern and display the target label for the images. Plot at least 9 clean images and display the target label.

Step 3: Create a poisoned test dataset, by using 20% of the total number of images of digit 2, and apply the backdoor pattern to create poisoned images for testing. Print the size of the test dataset and the number of poisoned images.

Estimated time: between 5 and 10 minutes.

Report (10 marks): (a) Plot the required figures in Steps 1 to 3. (b) Write between 5 and 10 sentences to explain the created training and test datasets. State how many poisoned images are used in the training dataset and how many poisoned images are used in the test dataset. State the total number of images in the training and test datasets.

Task 3: Implement a backdoor poisoning attack against the trained model.

Step 1: Train a poisoned model by finetuning the VGG model from Task 1 on the poisoned training dataset of images.

Evaluate the poisoned model on clean test images and report the classification accuracy. The classification accuracy on clean test images should be at least 90%, and the accuracy on the poisoned images should be also at least 90%. If these accuracies are not achieved, consider retraining the model either for fewer or more epochs.

Step 2: Plot at least 9 poisoned test images and show the target and predicted class label. Plot at least 9 clean test images and show the original and predicted class label. Plot at least 9 clean test images of digit 2 and show the original and predicted class label. Plot at least 9 clean test images of digit 5 and show the original and predicted class label.

Estimated time: between 5 and 20 minutes.

Report (20 marks): (a) Report the classification accuracy of the poisoned model on the clean and poisoned test images. For full marks, it is expected to report accuracy above 90%. (b) Plot the

required figures in Step 2. (c) Describe the results of the backdoor attack using between 5 and 10 sentences.

Part 2: Network Intrusion Detection Attacks (50 marks)

Dataset: We will use a portion of the UNSW-NB15 dataset for network intrusion detection. The dataset contains 44 features extracted from network data packet records (PCAP files), which include duration of the connection, protocol type, data bytes sent from source to destination, number of failed logins, and similar information. The dataset has 175K records in a training set and 82K records in a test set, which are provided in the assignment folder. The label for each record indicates whether it is normal traffic or an attack. The lecture notes on Network Intrusion Detection provide basic information about PCAP files, as well as more detailed information can be found at this [link](#).

Task 1: Load the dataset and perform basic data exploration, categorical feature encoding, and data normalization, similar to the code provided in the `Intrusion_Detection_Attack` notebook posted in the `Intrusion_Detection` folder on Canvas.

Estimated time: between 5 and 10 minutes.

Report (10 marks): (a) Provide the corresponding plots for the above steps. (b) Briefly describe the dataset. Is the dataset imbalanced? Are there attacks in the test datasets that are not present in the training dataset?

Task 2: Step 1. Train the following conventional ML models to classify the network connections into normal and attack classes: Logistic Regression, Decision Trees, Random Forest, k-Nearest Neighbors, naïve Bayes, and Support Vector Machines. It is not expected to finetune these models, but feel free to perform some finetuning if you wish to. Report the classification accuracies on the test dataset by the different models.

Step 2. Train a deep-learning model for classification of the network connections into normal and attack classes. You can consider a network with several fully-connected layers and dropout layers, or you can consider other types of layers if you wish. Report the classification accuracy on the test dataset, and display the learning curves. For full marks, it is expected to obtain test accuracy greater than 85%.

Estimated run time: between 10 and 20 minutes.

Report (15 marks): (a) Report the classification accuracies for the models. The accuracies for most conventional models should be around 80 to 85%. For the NN, accuracy over 85% is required for full marks. (b) Elaborate on the performance of the different models and compare the accuracies.

Task 3: Step 1. Implement a non-targeted white-box FGSM evasion attack against the deep learning model from Task 2. For creating the adversarial samples, use a random set of 1,000 records from the test dataset. Apply the following perturbation magnitudes: $\epsilon = [0.01, 0.025, 0.05, 0.1, 0.2, 0.3, 0.5]$.

Step 2. Implement a non-targeted white-box JSMA (Jacobian Saliency Mapp Attack) against the deep learning model from Task 2. In the ART library, JSMA is implemented using

[SaliencyMapMethod](#). For creating the adversarial samples, use the same random set of 1,000 records. Apply three JSMA attacks with different values of the parameters for the function in the ART library. Discuss the results and compare the performance to the FGSM.

Step 3. Implement a non-targeted white-box FGSM against the Logistic Regression model from Task 2. The following [notebook](#) in the ART toolbox provides an example of implementing attacks against Logistic Regression. For creating the adversarial samples, use the same random set of 1,000 records. Apply the following perturbation magnitudes: $\epsilon = [0.01, 0.025, 0.05, 0.1, 0.2, 0.3, 0.5]$. Discuss the results and compare the performance to the attacks against the deep learning model from Steps 1 and 2.

Estimated run time: between 5 and 15 minutes.

Report (25 marks): (a) Report the classification accuracies for the different attacks. (b) Explain the differences in the effectiveness of the attacks against the different models.

Submission documents:

The assignment documents are submitted on Canvas.

1. Submit all your codes as Jupyter Notebooks. Please try to comment your codes extensively, introduce the names of all used variables, avoid one-letter variables, etc., to improve the readability of the codes. For instance, for this assignment you can submit two Jupyter Notebooks with the codes. You don't need to submit the dataset on Canvas.
2. Prepare a brief report with tables, graphs, and results: the report can be prepared either as a separate MS Word/PDF file, or it can be integrated into your Jupyter Notebooks by typing your analysis directly into text cells in the Jupyter Notebooks.